

A.Y. 2025-2026

Master Degree in Bionics Engineering

Course of Principles of Bionics and Biorobotics Engineering

Lesson title:

Exam modality

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THE BIORBOTICS
INSTITUTE



Scuola Superiore
Sant'Anna



Exam modality

Principles of Bionics and Biorobotics Engineering

Exam dates

January 07, 2026

January 26, 2026

February 16, 2026

<https://www.ing.unipi.it/it/studenti/calendario-esami>

Oral exam on the topics presented during the course.

Students will be also asked to «solve» rather unstructured problems in which creative bionic design principles should be used.



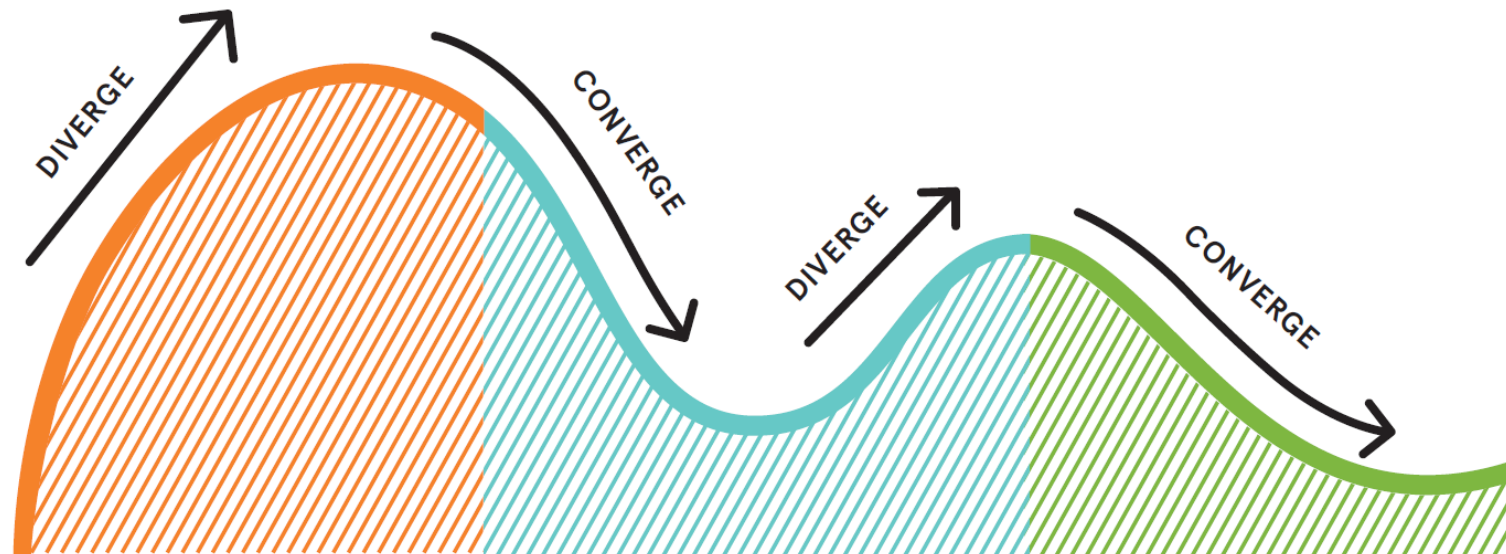
Exam modality

These projects will be the first step to accomplish your exam!



Creative Bionics Design

Before the individual oral exam (covering all the topics discussed during the course), you are asked to form groups of maximum 3 members. Each group of students must prepare a presentation of approximately 10 slides. The goal is to identify a problem in any context(industrial, medical, urban, environmental, space-related, or others) and propose an innovative solution by applying the principles of bionic design covered during the course.



Convergent thinking is an ideation mode which designers use to analyze, filter, evaluate, clarify and modify ideas they have generated in divergent thinking. They use analytical, vertical and linear thinking to find novel and useful ideas, understand the design space possibilities and get closer to potential solutions.



Creative Bionics Design

Presentations should follow this structure:

1.Problem Identification & Context

Introduce the selected problem, explaining the context and why it is relevant.

2.State of the Art

Summarize existing solutions, approaches, or technologies related to the problem.

3.Gap Analysis

Highlight what is missing or insufficient in current solutions.

4.Innovative Idea (Your Proposal)

Present your original bionic design concept inspired by biological systems.

5.Concept Development / Prototype Storytelling

Provide a technical description of how your conceptual prototype would work.

(Note: This is not a real physical prototype. Use storytelling to demonstrate creativity, originality, and your engineering mindset)

6.Feasibility Assessment

Discuss technical and economic feasibility.

7.Potential Impact & Applications

Explain the broader implications and possible real-world applications of your solution.



Examples of previous themes addressed

Cargo robot

Working animals were kept by humans and trained to perform tasks (e.g. transportation, riding, packing, etc.), especially in the past, although they are still largely used in many regions worldwide. Ethical, economic and technological issues make the use of beasts of burden no longer compatible with our era. It's bionics time!

Design a robot moving easily in highly unstructured environments (e.g. forests, deserts, swamps, etc.), able to carry a cargo.

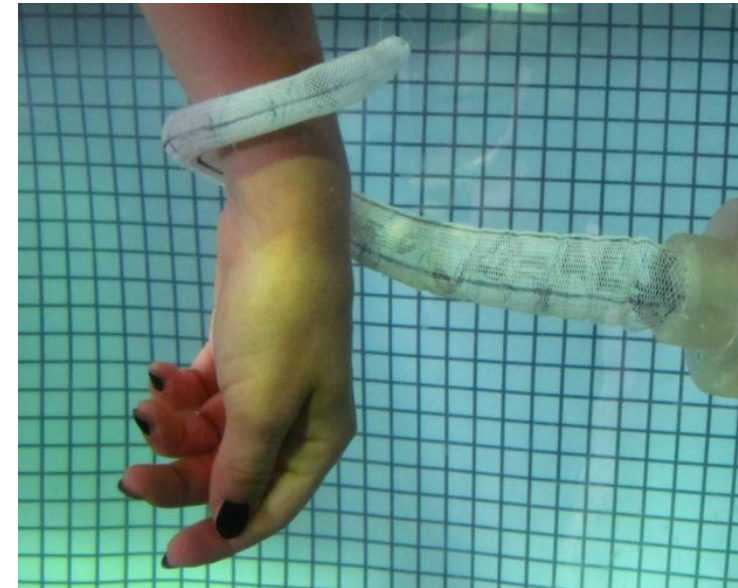
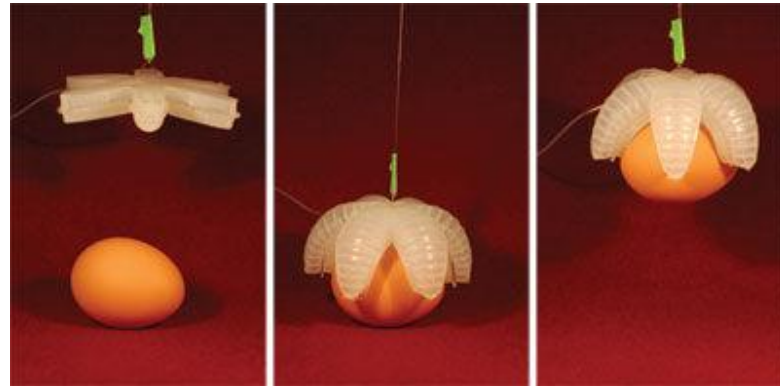
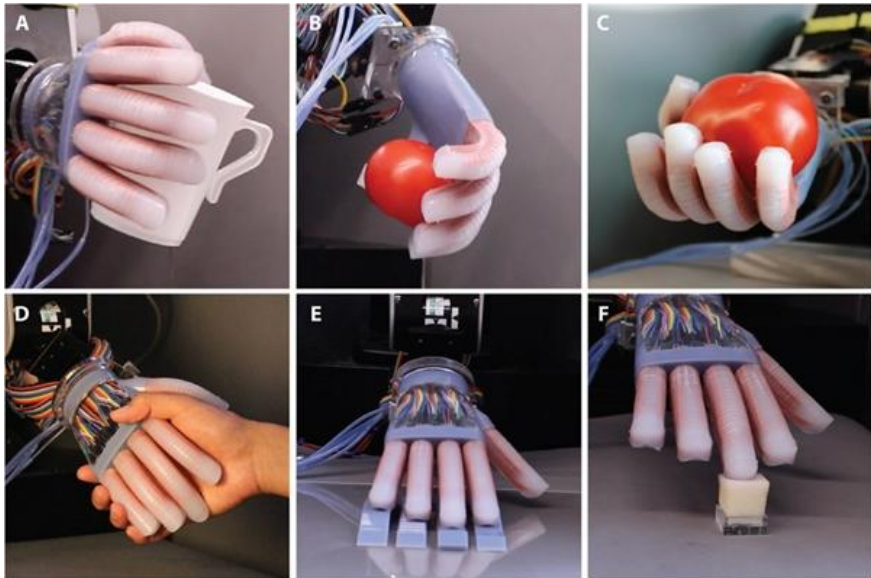


Examples of previous themes addressed

Soft touch

Recent advances in soft and smart materials allowed the designing of mechanisms for varying the material characteristics such as stiffness, elasticity, and surface properties

Design a custom-made soft actuator, inspired by animal structures, to manipulate delicate objects.



Examples of previous themes addressed

Swimming against the current

A broken pipeline is spilling oil into a stream and you have to find the exact point of the oil source along the stream.

Design a bioinspired artefact able to face water current, obstacles and endowed with appropriate sensors for navigation and detection.

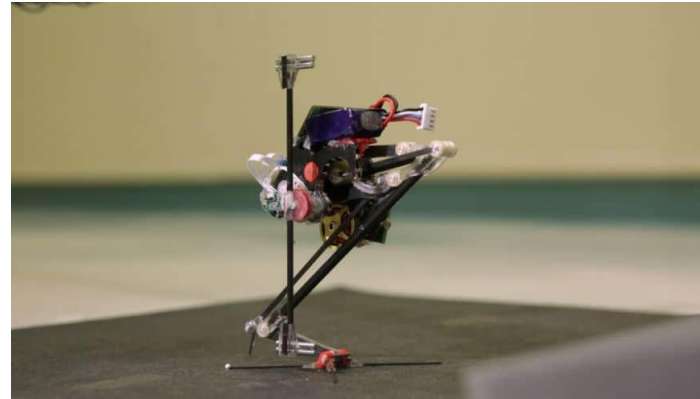


Examples of previous themes addressed

Distance measuring jumping robot

An agent has to move in a minefield with mines position marked by flags. Jumping on a specific target is crucial to accomplish the mission.

Explore the role of spatial cues and distance measurements as well as jumping mechanisms in animals to design a jumping robot able to calibrate the jump according to the target to reach.



Examples of previous themes addressed

Underground robot

Tuscany, and in general Italy, has numerous ancient buildings and other infrastructure that often are defaced/damaged by modern installations and constructions.

Design a bionic agent, inspired by telluric animals (earthworms, mole crickets, moles, etc.), that can help install underground pipes and cables by navigating under/around existing infrastructure.



How students faced challenges

Robadillo

Design of a cargo robot able to move easily in highly unstructured environments

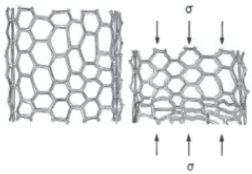
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Giorgio Bondi
Matteo Lo Preti
Federico Masiero

Scientific Basis

Cellular (honeycomb-like) structure

It allows to obtain high material toughness, by minimizing the weight.



By analyzing the stress and the torque:

$$\sigma = \frac{MR}{I} \approx \frac{M}{\pi R^2 t}$$

σ : stress
 M : bending momentum
 R : external radius
 I : inertia moment

$$\tau = \frac{TR}{J} \approx \frac{T}{2\pi R^2 t}$$

τ : shear stress
 T : torque
 J : polar moment of inertia

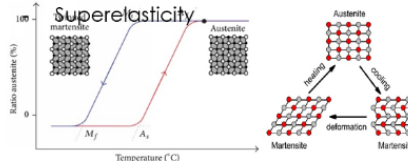
we can conclude that both I and J increase, without increasing the weight.

Nickel Titanium

Nitinol is a shape memory alloy with enhanced wear resistance and ductility.

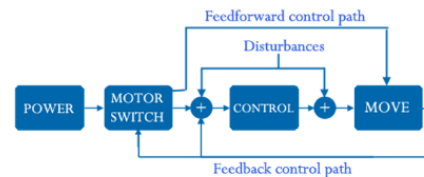
Material Properties	
Density	6.45 g/cm ³
Electrical Resistivity	82×10 ⁻⁶ Ω·cm (A) 76×10 ⁻⁶ Ω·cm (M)
Coefficient of thermal expansion	11×10 ⁻⁶ /°C (A) 6.6×10 ⁻⁶ /°C (M)
Elastic Modulus	75–83 GPa (A) 28–40 GPa (M)
Yield strength	195–690 MPa (A) 70–140 MPa (M)

- Twinned martensite structure
- Instantaneous reversible transformation



Caterpillar Inspired Roll Control

Active rolling → self-sufficient energy
Recoil-roll mechanism → speed and efficiency



Feedforward rhythmic component → CPGs and nonlinear robot dynamics create the desired rolling gait trajectory.
Feedback method → a nonlinear integral sliding mode (ISMC) gives stability and robustness in the presence of external disturbances.

BIOINSPIRED ROBOT



Concept Description

A cargo robot is a machine able to collect information, interact with the environment and aimed at carrying materials.

- Biomimetic system
- Two types of locomotion
- Compliant elastic structure
- Different possible control systems
- Two kinds of motors
- Self actuated folding mechanism
- Pouch-like unit
- Low power electronics
- Four-legged system (2.0 × 0.4 × 0.5)m
- Wheel: $d = 0.7\text{m}$, $l = 0.5\text{m}$
- Load capacity: 80L

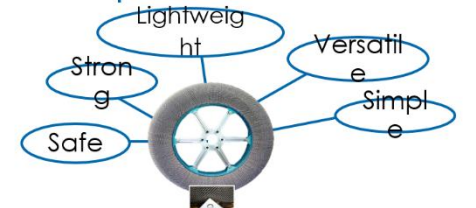
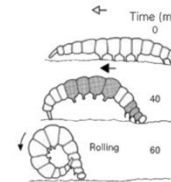
NATURE



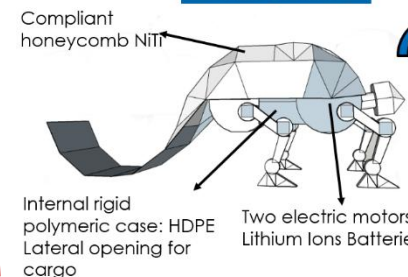
Technical solutions adopted

GoQBot: a caterpillar-inspired soft-bodied rolling robot

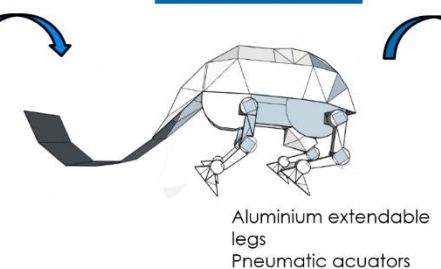
- Fast change of body conformation (100ms).
- Ballistic roll is the best energy efficient mode of locomotion on flat ground.
- Morphological computation.
- Instantaneous muscle recruitment



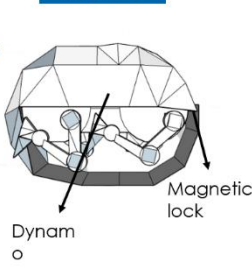
STANCE PHASE



PROPULSIVE PHASE



ROLLING



How students faced challenges

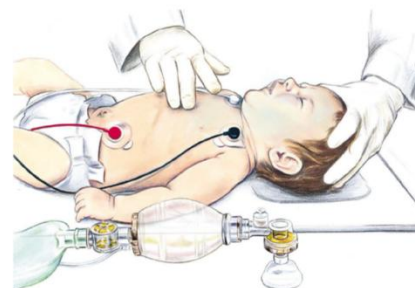
Newborns' Ca.Re. Machine

Design of a soft actuator device for newborns' cardiac resuscitation

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Anna Astolfi
Elena Manferlotti

Two ways of chest compression



Two fingers

Which one is better?



Two thumbs

Compression pressure: chest reduction of 1/3 (≈ 4 cm)

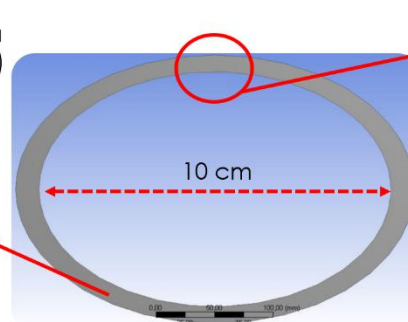
Compression to inflation ratio = 3:1

Normal HR about $120 - 150 \text{ min}^{-1}$

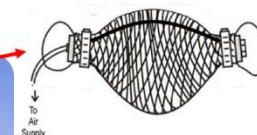
New Idea: Soft robot for cardiac massage and temperature keeper

Hypothermia
($36,5-37,5^\circ\text{C}$)

Liquid Metal
Embedded
Elastomer



CO2 pressure pump



Constrained
McKibben
Actuator

Chest compression

How students faced challenges

Leak Freak

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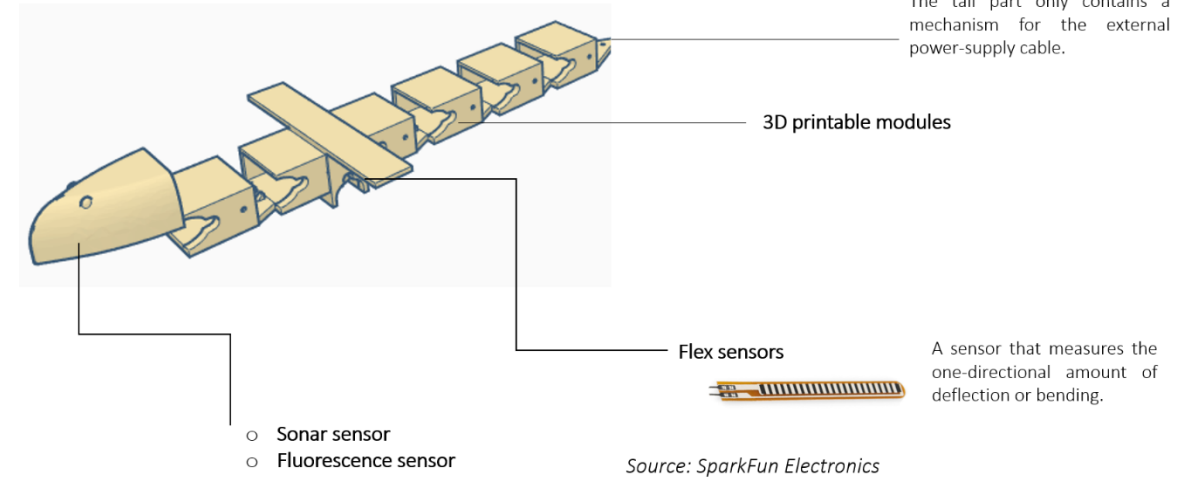
Conrad Helminger

Mariana Pereira

Pedro Dias

24.01.2018

Concept Description



Locomotion

Locomotion of an underwater snake



Source: BlueWorldTV

Snake-inspired diving robot (5 joint modules)



Source: Liljebäck et al.: Mamba, 2014.

Characteristics of a 10 module configuration:

- 3 kg in weight
- Simulated power consumption of 40 W
- At a velocity of 0.9 m/s

How students faced challenges

FROG-BALL

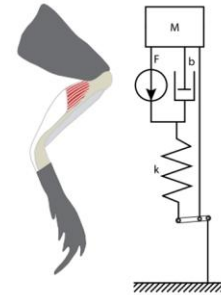
Design of a distance-measuring jumping robot

Federica G. Cornacchia Loizzo - 572874
Arturo Castillo Ugalde - 572892
Jacopo Quaglierini - 524579

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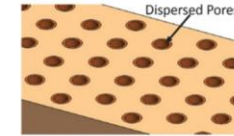
Scientific Basis

Catapult-like jumping



Many frogs store elastic energy in their tendons to obtain an output power greater than the one possible with muscles alone.[4]

Tubular structure



The arrays of long aligned pores (tubules) within the bulk material decrease the stiffness but absorb energy during compression.

$$\sigma_c = \frac{E_m (1 - V_p^{\frac{1}{3}}) \epsilon_{mf}}{v_m}$$

$$E = E_0 (1 - V_p^2)$$

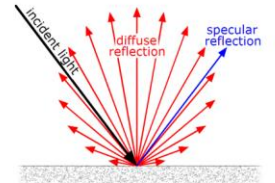
Photovoltaic effect

Light shining on a particular material causes the emission of electrons, generating a voltage and an electric current in it.

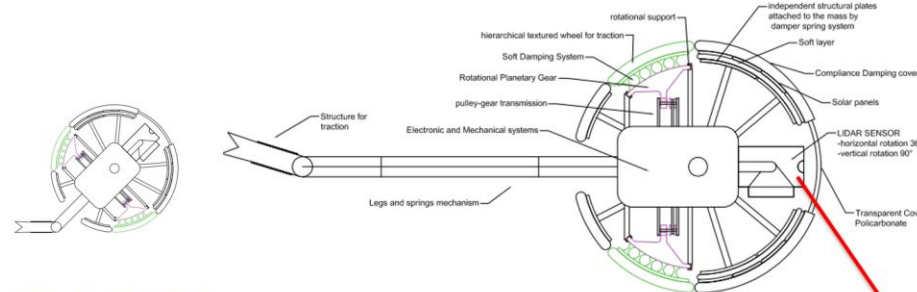


Backscattering

Waves are reflected back to the direction from which they came, but with diffuse reflection due to scattering, as opposed to specular reflection like a mirror.

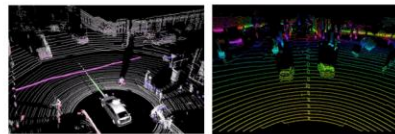


Technical solutions adopted: LIDAR and catapult jump

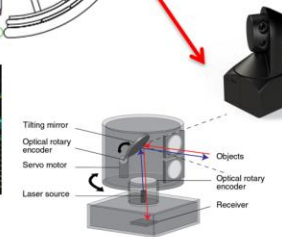


LIDAR SENSING STRATEGY

- High precision 3D mapping
- Sensor fusion (RADAR, cameras, GPS)
- Detection of a wide range of materials (metals, rock, rain)
- Versatility (slopes, wind speed, T, P, humidity estimations via reflectance measurements)



GPS receiver and IMU are incorporated into the sensor to get relative and absolute orientation

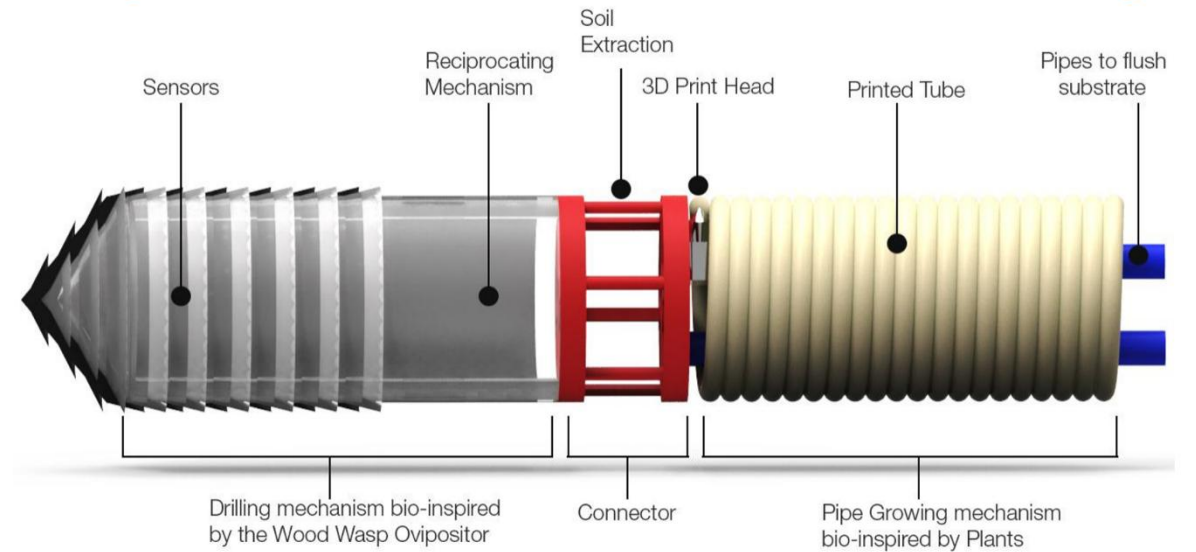


How students faced challenges

Concept - 3D model

G-Pipe

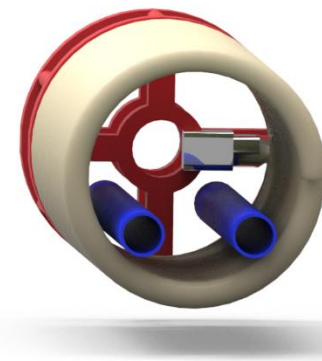
Akshat Wahi
Philip Maus
Ingibjörg Tómasdóttir



Digging/flushing out with water



Wood Wasp



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Oral exam on the topics presented during the course.

All slides available in the shared folder.

Students will be also asked to find and «solve» rather unstructured problems in which creative bionic design principles should be used, starting from the concepts elaborated during the group activities.

